

CITE: Class-to-Instance Text-Guided Image Editing

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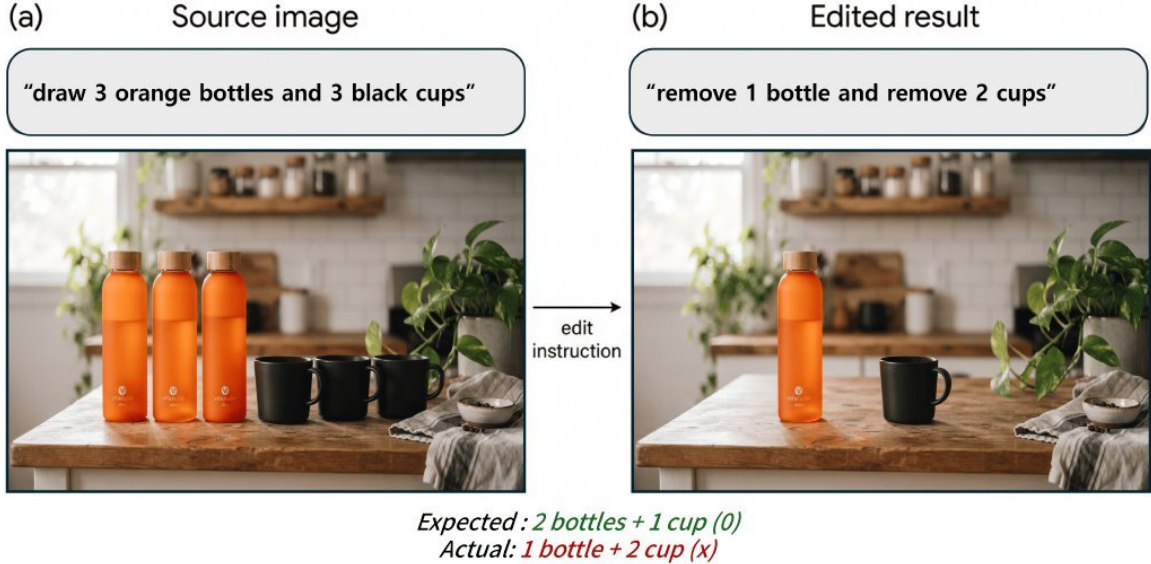


Figure 1: A failure case of instance-level control in text-guided image editing. This scenario was created with Nano Banana 2.

Abstract

Recent text-guided image editors can perform diverse visual editing tasks from natural-language instructions. However, these models are mainly trained to understand object classes, while images are composed of individual object instances. This creates a class-to-instance control gap. An editor may know what a dog or a cup is. However, it may still fail to decide which instances should be edited and which should be preserved. To address this gap, we propose Class-to-Instance Text-Guided Image Editing (CITE), a training-free framework for instance-level editing. Given an instruction, CITE compares the number of instances that match the instruction with the requested count, and classifies the instruction as uniquely-determined, under-determined, or over-determined. CITE then grounds target and protect instances in the image, constructs an instance-aware proxy condition, and runs a pretrained text-guided image editor with a proxy-refinement prompt and per-step source-latent composition. The proxy condition tells the editor where the edit should happen, while latent composition preserves non-edit regions during denoising. We evaluate CITE on three editing operations, removal, recoloring, and resizing, measuring whether the requested instances are edited correctly while non-target objects remain unchanged. Across these settings, CITE improves instance-level preservation and control while maintaining competitive target-edit success rel-

ative to raw editing.

1 Introduction

Consider the example in Figure 1. A user first creates a scene with three bottles and three cups, and then asks the editor to “remove one bottle and two cups.” The correct result should contain two bottles and one cup. However, current editors often return one bottle and one cup. This is not a failure of class recognition. The model recognizes bottles and cups correctly. The failure is that the model receives a class-level instruction but must execute an instance-level operation, and it has no mechanism to decide which instances to edit and which to preserve.

We call this problem the *class-to-instance control gap*. It is the failure in which an editor model understands what class should be edited, but cannot reliably determine which instances should be edited, and which instances should remain unchanged.

This paper proposes Class-to-Instance Text-Guided Image Editing (CITE), a training-free framework that turns a pretrained text-guided image editor into an instance-aware editor. CITE consists of three steps. First, CITE builds an instance inventory of the source image and classifies the instruction by a determinacy rule: an instruction is uniquely-determined when

the instances matching the instruction exactly match the requested count, under-determined when more instances match than requested, and over-determined when fewer instances exist than requested. Second, CITE selects target and protect instances from the inventory and constructs a proxy condition in which the selected targets have already been edited roughly according to the requested operation. Third, during denoising, CITE applies per-step source-latent composition, so the editor rerenders only the edit region while all non-edit regions follow the source image.

The key insight behind CITE is that instance selection and edit rendering are two separate problems that a class-level prompt forces the editor to solve at once. CITE takes the selection problem out of the prompt. Once target and protect instances are grounded in the image, the decision can be written directly into the editor’s inputs. The pretrained editor is then left with the one task it is good at, rendering a locally edited region naturally, while the instance-level decision is handled where it can be made reliably. As a result, CITE applies the edit to the selected instances and preserves the remaining ones. Our contributions are summarized as follows:

- We identify and formalize the class-to-instance control gap, a failure in which current text-guided image editing models follow class-level instructions but cannot control instance-level edits.
- We propose a determinacy rule that classifies an edit instruction as uniquely-determined, under-determined, or over-determined, based on the relation between the class-level request and the actual instances in the image.
- We propose a training-free, instance-aware editing framework that combines proxy conditioning with per-step source-latent composition, and validate it on instance-level removal, recoloring, and resizing.

2 Related Work

Method	Instruction-based	Auto mask	Instance-level	Background preserv.	Training-free
InstructPix2Pix	✓	✗	✗	△	✗
DiffEdit	△	✓	△	✓	✓
Prompt-to-Prompt	✗	✗	△	△	✓
Blended Latent Diff.	✗	✗	✗	✓	✓
Specify and Edit	✓	✗	△	△	✓
CITE (Ours)	✓	✓	✓	✓	✓

Table 1: Comparison with representative text-guided editing methods. ✓ supported, ✗ not, △ partial.

Table 1 summarizes the functional distinction between CITE and representative text-guided editing methods.

Text-guided image editing. Text-guided image editing has evolved from diffusion-based transformation to instruction-following editors. SDEdit edits images through a diffusion prior Meng et al. [2022], DiffEdit estimates edit masks from text-conditioned prediction differences Couairon et al. [2023], and InstructPix2Pix trains an instruction-following editor on synthesized image-editing pairs Brooks et al. [2023]. MagicBrush and Emu Edit further broaden instruction-guided editing benchmarks and tasks Zhang et al. [2023b], Sheynin et al. [2024].

Grounded and attention-guided editing. Several methods preserve layout or localize edits using attention, diffusion features, or masks. Prompt-to-Prompt, Null-text Inversion, Plug-and-Play, and MasaCtrl improve text-only or inversion-based preservation through attention and feature reuse Hertz et al. [2023], Mokady et al. [2023], Tumanyan et al. [2023], Cao et al. [2023]. Blended Latent Diffusion and Segment Anything show the value of explicit spatial constraints Avrahami et al. [2023], Kirillov et al. [2023]. However, these methods do not directly resolve a count-constrained target subset among repeated same-class instances.

Instruction ambiguity and determinacy. A few methods reason about ambiguous instructions. Specify and Edit decomposes ambiguous edits with a language model Iakovleva et al. [2025], Editor’s Choice studies abstract-intent editing Ventura et al. [2026], and object-count generation methods steer diffusion models toward requested counts Binyamin et al. [2025]. These works address what to do or how many objects to depict; CITE addresses which visible instances to edit.

3 Method

In each equation pair below, the first equation gives the general definition, and the following equation gives an instantiation using the running example in Figure 2.

3.1 Problem Setup and Notation

Let I be the source image and let x be the natural-language edit instruction.

$$\begin{aligned} I &= \text{image with three bottles and three cups} \\ x &= \text{“remove one bottle and two cups”} \end{aligned} \quad (1)$$

CITE first constructs an instance inventory:

$$\Omega = \{o_i\}_{i=0}^{n-1}. \quad (2)$$

$$\Omega = \{o_{\text{bt},0}, o_{\text{bt},1}, o_{\text{bt},2}, o_{\text{cup},0}, o_{\text{cup},1}, o_{\text{cup},2}\}. \quad (3)$$

Ω is the set of object instances detected in the source image. n is the number of instances in Ω . o_i is the i -th instance entry in the inventory. Each o_i is represented by a class label, optional attributes, and an instance mask m_i .

CITE then decomposes the instruction into individual edit requests:

$$\mathcal{R} = \{r_k\}_{k=1}^K, \quad r_k = (\rho_k, q_k, a_k). \quad (4)$$

$$\mathcal{R} = \{r_1 = (\text{bottle}, 1, \text{remove}), r_2 = (\text{cup}, 2, \text{remove})\}. \quad (5)$$

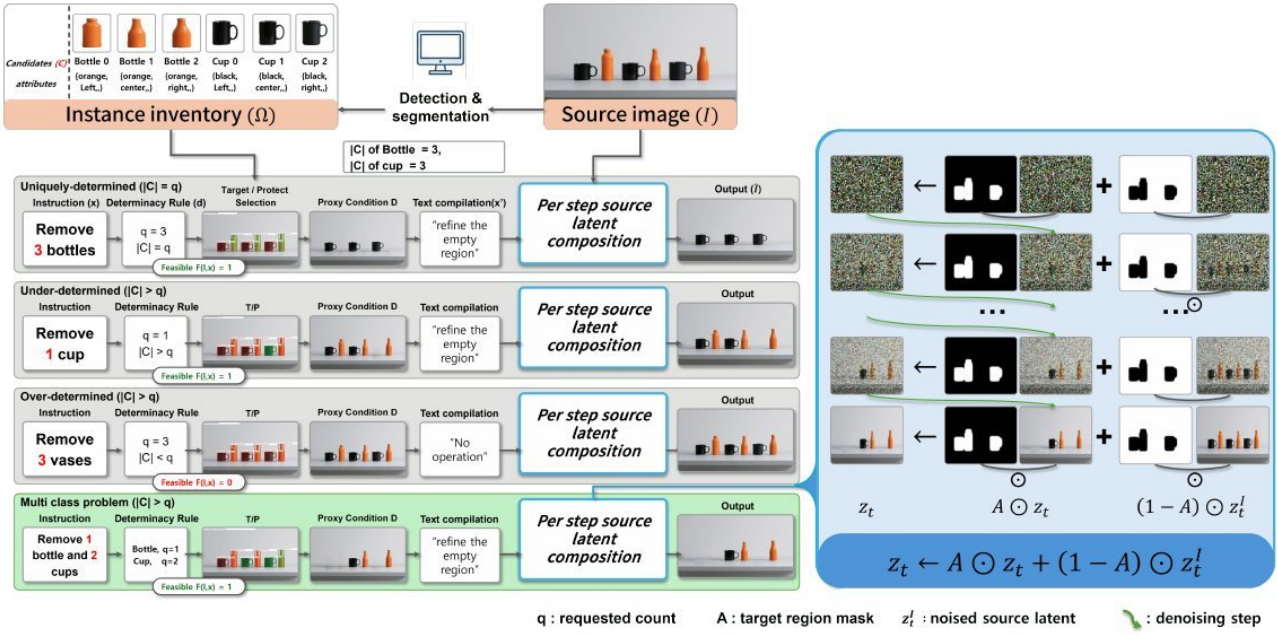


Figure 2: Overview of CITE. Given a source image and a class-level edit instruction, CITE builds an instance inventory, classifies the instruction by determinacy, selects target and protect instances, constructs an instance-aware proxy condition, and refines the proxy with per-step source-latent composition.

Here, $\mathcal{R}$ is the set of individual edit requests and K is the number of requests. Each request $r_k = (\rho_k, q_k, a_k)$ contains an object condition ρ_k , a requested count q_k , and an edit operation a_k , where ρ_k specifies the target class or attribute-conditioned category and a_k specifies the operation to perform. For each request r_k , CITE constructs a candidate set:

$$C_k = \{o_i \in \Omega \mid o_i \text{ matches } \rho_k\}. \quad (6)$$

$$C_1 = \{o_{bt,0}, o_{bt,1}, o_{bt,2}\}, \quad C_2 = \{o_{cup,0}, o_{cup,1}, o_{cup,2}\}. \quad (7)$$

3.2 Determinacy and Instance Sets

For each individual edit request r_k , CITE compares the requested count q_k with the number of available candidates $|C_k|$:

$$d_k = \begin{cases} \text{Uniquely-determined,} & |C_k| = q_k, \\ \text{Under-determined,} & |C_k| > q_k, \\ \text{Over-determined,} & |C_k| < q_k. \end{cases} \quad (8)$$

$$|C_1| = 3 > q_1 = 1, \quad |C_2| = 3 > q_2 = 2, \\ d_1 = d_2 = \text{Under-determined.} \quad (9)$$

CITE defines instruction-level feasibility as

$$F(I, x) = \prod_{k=1}^K \mathbf{1}[|C_k| \geq q_k]. \quad (10)$$

$$F(I, x) = \mathbf{1}[3 \geq 1] \cdot \mathbf{1}[3 \geq 2] = 1. \quad (11)$$

The bottle request is feasible because three bottles are available for one requested removal, and the cup request is feasible because three cups are available for two requested removals. Therefore, the full instruction is feasible.

CITE selects target instances only when the full instruction is feasible:

$$T_k = \begin{cases} C_k, & F(I, x) = 1, d_k = \text{unique}, \\ S(C_k, q_k), & F(I, x) = 1, d_k = \text{under}, \\ \emptyset, & F(I, x) = 0. \end{cases} \quad (12)$$

$$T_1 = \{o_{bt,0}\}, \quad T_2 = \{o_{cup,0}, o_{cup,1}\}. \quad (13)$$

S is a selector that ranks candidates by instance mask area and returns the q_k largest ones. Any selection rule can be substituted; the choice affects which valid subset is edited, not whether the requested count is met. The final target and protect sets are

$$T = \bigcup_{k=1}^K T_k, \quad P_{\text{protect}} = \Omega \setminus T. \quad (14)$$

$T = \{o_{bt,0}, o_{cup,0}, o_{cup,1}\}, \quad P_{\text{protect}} = \{o_{bt,1}, o_{bt,2}, o_{cup,2}\}.$ (15)
 P_{protect} contains the non-target instances that should be preserved. For infeasible instructions, CITE sets

$$T = \emptyset, \quad P_{\text{protect}} = \Omega. \quad (16)$$

$$|C_1| = 1 < q_1 = 2 \Rightarrow F(I, x) = 0, \quad T = \emptyset. \quad (17)$$

In this case, CITE applies a no-operation policy and preserves the source image.

3.3 Proxy Image Condition

CITE constructs a proxy image condition D from the source image as a rough draft of the intended instance-level edit. For each request r_k , CITE computes an edit-support mask $M_{E,k}$:

$$M_{E,k} = \Gamma_{a_k}(I, T_k, \Omega). \quad (18)$$

$$\begin{aligned} M_{E,1} &= \Gamma_{\text{remove}}(I, \{o_{\text{bt},0}\}, \Omega), \\ M_{E,2} &= \Gamma_{\text{remove}}(I, \{o_{\text{cup},0}, o_{\text{cup},1}\}, \Omega). \end{aligned} \quad (19)$$

Here, Γ_{a_k} is a deterministic, action-dependent mask operator returning the region the editor may modify: removal widens the target masks to cover contact and shadow cues, recoloring keeps them as they are, and resizing unions them with the scaled target regions. Protected-instance cores are always excluded. The supplementary sections on operation-specific proxy construction and mask construction provide the full definitions. The full edit-support mask is

$$M_E = \bigcup_{k=1}^K M_{E,k}. \quad (20)$$

$$M_E = M_{E,1} \cup M_{E,2}. \quad (21)$$

CITE constructs the proxy condition using the selected targets and the edit-support mask:

$$D = \begin{cases} \Phi_{\mathcal{R}}(I, T, M_E, \Omega), & F(I, x) = 1, \\ I, & F(I, x) = 0. \end{cases} \quad (22)$$

$$F(I, x) = 1 \Rightarrow D = \Phi_{\mathcal{R}}(I, T, M_E, \Omega). \quad (23)$$

Here, $\Phi_{\mathcal{R}}$ is a deterministic image operator: removal fills the target regions with background, recoloring shifts their color toward the request, and resizing fills them and pastes back the scaled target.

3.4 Compiled Text Condition

CITE compiles the original instruction x into a compiled text condition x' :

$$x' = \begin{cases} x'_{\text{refine}}(x, T, P_{\text{protect}}), & F(I, x) = 1, \\ x'_{\text{preserve}}, & F(I, x) = 0. \end{cases} \quad (24)$$

$$F(I, x) = 1 \Rightarrow x' = x'_{\text{refine}}(x, T, P_{\text{protect}}). \quad (25)$$

$$\begin{aligned} x &= \text{"Remove one bottle and two cups"}, \\ x'_{\text{refine}} &= \text{"Treat the proxy image as an edited draft.} \\ &\quad \text{Refine the edited regions according to the instruction.} \\ &\quad \text{Preserve all visible non-target objects."} \end{aligned} \quad (26)$$

3.5 Per-Step Source-Latent Composition

CITE preserves non-edit regions by composing a conditioned denoising latent with a source latent at each step. Let z_t denote the conditioned denoising latent at timestep t under (D, x') . CITE constructs the source latent z_t^I from the source image:

$$z_t^I = \alpha_t \text{Enc}(I) + \sigma_t \epsilon, \quad t \in \mathcal{S}. \quad (27)$$

Here, Enc is the VAE encoder, α_t and σ_t are scheduler coefficients, and $\mathcal{S}$ is the set of denoising timesteps. The noise sample ϵ is shared with the conditioned denoising trajectory, so z_t^I is re-noised to the corresponding timestep. CITE maps the edit-support mask M_E to the latent resolution:

$$A = \mathcal{B}(M_E). \quad (28)$$

$$A = \mathcal{B}(M_{E,1} \cup M_{E,2}). \quad (29)$$

Here, A denotes the edit region in latent space, $\mathcal{B}$ binarizes, downsamples, dilates, and feathers M_E to obtain a smooth latent-space mask. Its fixed parameters are reported in the supplementary hyperparameter section. At each denoising step, CITE applies source-latent composition:

$$z_t \leftarrow A \odot z_t + (1 - A) \odot z_t^I, \quad t \in \mathcal{S}. \quad (30)$$

The update uses the conditioned denoising latent inside the edit-support region and the source latent outside it. Following BLD Avrahami et al. [2023], the updated z_t is passed to the next denoising step. After the final denoising step, CITE decodes z_0 with the VAE decoder Dec :

$$\hat{I} = \text{Dec}(z_0). \quad (31)$$

4 Experiments

4.1 Experimental Setup

We use Qwen-Image-Edit 20B as the text-guided image editor Wu et al. [2025], and construct instance inventories with YOLO11x Ultralytics [2024] and MobileSAM Zhang et al. [2023a]. All experiments run on a single NVIDIA RTX PRO 6000 GPU. The Baseline uses the same pretrained editor as CITE. It receives the source image and the raw instruction. Evaluation uses two independent instance detectors, Mask2Former Cheng et al. [2022] and Mask R-CNN He et al. [2017], with a shared DINOv2 matcher Oquab et al. [2024].

4.2 Benchmark

We evaluate CITE on three operations: removal, recoloring, and resizing. Each operation uses instructions that specify an object description, e.g., "remove two bottles" or "make one cup blue." The benchmark includes both single-class instructions and multi-class instructions such as "remove one bottle and one cat." The categories include vases, bottles, cups, chairs, cats, dogs, cars, and birds. All source images are generated with Qwen-Image to construct scenes with controlled object categories and instance counts. Our main benchmark covers the two feasible determinacy types, uniquely- and under-determined. Over-determined requests are infeasible and follow a no-operation policy, which is described in the supplementary no-operation section. Table 3 summarizes the benchmark.

4.3 Metrics

We propose Target Edit Success (TES) and Protect Preservation (PP). For each detector, DINOv2 features match each source instance o_i to an output instance $\hat{o}_i$, or to $\emptyset$ if no match is found. TES and PP are computed separately for each method and detector.

Operation	Determinacy	Method	Mask2Former		Mask R-CNN		Consensus			Source fidelity					Generation time	
			TES↑	PP↑	TES↑	PP↑	CITE Win	Baseline Win	Tie	LPIPS-W↓	LPIPS-P↓	DSim-W↓	DSim-P↓	CLIP-W↑	CLIP-P↑	Time (s)↓
Removal	Uniquely-determined	Baseline	0.711	0.616	0.695	0.576	76.8%	2.9%	20.3%	0.413	0.275	0.182	0.058	0.910	0.964	17.313
		CITE	0.989	0.881	0.997	0.873				0.146	0.001	0.199	0.000	0.920	1.000	17.506
	Under-determined	Baseline	0.224	0.447	0.210	0.434	93.0%	2.0%	5.0%	0.411	0.294	0.186	0.085	0.907	0.951	17.319
		CITE	0.904	0.903	0.905	0.895				0.119	0.001	0.103	0.000	0.964	1.000	17.508
Recolor	Uniquely-determined	Baseline	0.797	0.417	0.802	0.392	87.4%	1.8%	10.8%	0.473	0.433	0.166	0.107	0.935	0.945	17.335
		CITE	0.959	0.983	0.966	0.975				0.022	0.001	0.060	0.000	0.985	1.000	17.530
	Under-determined	Baseline	0.446	0.389	0.453	0.376	93.0%	1.6%	5.4%	0.469	0.430	0.161	0.118	0.941	0.946	17.343
		CITE	0.969	0.981	0.966	0.979				0.023	0.001	0.049	0.000	0.988	1.000	17.538
Resize	Uniquely-determined	Baseline	0.605	0.542	0.619	0.525	86.8%	6.1%	7.1%	0.434	0.424	0.118	0.121	0.945	0.938	17.339
		CITE	0.928	0.948	0.947	0.943				0.062	0.046	0.028	0.020	0.991	0.993	17.533
	Under-determined	Baseline	0.246	0.523	0.239	0.515	94.6%	0.8%	4.6%	0.423	0.413	0.095	0.096	0.953	0.950	17.341
		CITE	0.963	0.952	0.972	0.950				0.061	0.040	0.022	0.014	0.994	0.995	17.534

Table 2: Main comparison using the same pretrained editor. Metrics are computed independently with Mask2Former and Mask R-CNN. Consensus wins use mean TES across detectors, with mean PP as the tie-breaker. Source-fidelity columns report whole-image (W) and preserve-view (P) LPIPS, DreamSim, and CLIP image similarity. Generation time reports the mean per-task runtime in seconds.

Operation	Uniquely-determined	Under-determined	Total
Removal	375	625	1000
Recolor	375	625	1000
Resize	375	625	1000
Total	1125	1875	3000

Table 3: Benchmark composition by determinacy.

Target Edit Success (TES). For each edit request k , let $b_{i,k} = 1$ if candidate instance o_i satisfies the operation-specific success criterion, and $b_{i,k} = 0$ otherwise. The criterion checks disappearance for removal, movement toward the requested color for recoloring, and the requested scale change for resizing. TES requires every edit request to match the requested count exactly:

$$\text{TES} = \prod_{k=1}^K \mathbf{1} \left[\sum_{o_i \in C_k} b_{i,k} = q_k \right]. \quad (32)$$

For under-determined requests, the evaluator ranks the operation evidence and selects the top- q_k edited candidates for each method independently. Thus, each method is credited for any valid size- q_k subset.

Protect Preservation (PP). Let $\tilde{T}_k$ be the top- q_k edited candidates for request k . For each method, the protect set is

$$P = \Omega \setminus \bigcup_{k=1}^K \tilde{T}_k. \quad (33)$$

A protected instance receives $g_i = 1$ if its matched output preserves its identity, location, scale, and color, and $g_i = 0$ otherwise. PP is defined as

$$\text{PP} = \frac{\sum_{o_i \in P} g_i}{|P| + |U|}, \quad (34)$$

where U is the set of unmatched new output instances. As auxiliary source-fidelity checks, we also report LPIPS Zhang

et al. [2018], DreamSim Fu et al. [2023], and CLIP image similarity Radford et al. [2021] on the whole image and on a preserve view that masks out the edit-support region. The complete matching rules, operation-specific criteria, source-fidelity checks, and fixed thresholds are provided in the supplementary metric and hyperparameter sections.

4.4 Main Results

Table 2 reports the feasible-task comparison by operation and determinacy. We examine whether the requested count and operation are satisfied while same-class survivors, other objects, and the background remain stable. This directly tests instance selection and preservation. CITE improves TES and PP across all operation-determinacy settings under both detectors. The largest gains occur for under-determined requests. Source fidelity shows the same trend: whole-image fidelity (W) compares the full output with the source, while preserve-view fidelity (P) evaluates only regions outside the edit support. CITE improves both, indicating that edits remain localized.

4.5 Qualitative Analysis

Figure 3 compares the Baseline and CITE across removal, resizing, and recoloring. The Baseline may remove too many objects, alter nearby objects during resizing, or recolor multiple candidates, whereas CITE more consistently applies the requested edit to the specified instances while preserving the remaining scene.

4.6 Instance-Aware Prompt Baseline

The Baseline in Table 2 receives the raw class-level instruction. To test whether prompting alone can address the class-to-instance control gap, we introduce an instance-aware prompt baseline that provides the source image together with CITE’s target and protect decisions in text. On the 300-task subset in Table 4, this baseline improves overall performance but still remains well below CITE. Figure 4 shows that even when the target is explicitly specified, the editor may still modify an addi-



Figure 3: Qualitative comparison. CITE edits the requested number of instances while preserving the remaining objects and background regions.

Method	Instance-level text	Mask2Former		Mask R-CNN		Average	
		TES $\uparrow$	PP $\uparrow$	TES $\uparrow$	PP $\uparrow$	TES $\uparrow$	PP $\uparrow$
Baseline (raw)	$\times$	0.507	0.489	0.516	0.477	0.511	0.483
Baseline (instance-aware prompt)	$\checkmark$	0.556	0.702	0.557	0.673	0.557	0.687
CITE	$\checkmark$	0.965	0.941	0.958	0.943	0.962	0.942

Table 4: Instance-aware baseline on a 300-task subset.



Baseline prompt: "change 1 bottle to purple"

Instance-aware prompt: "Change the bottle at the left area to purple. Keep the instance shape, pose, scale, and location unchanged. Keep all brown dogs, orange vases, red bottles, and red dogs unchanged. Edit exactly one selected target object. Keep every non-target instance and the rest of the scene unchanged."

Figure 4: Comparison with the instance-aware prompt baseline

tional instance, indicating that instance-level prompting alone is insufficient.

4.7 Evaluation on Real Images

The main benchmark uses generated source scenes, which give controlled instance counts but share a generator family with the editor. To test whether the class-to-instance control gap persists outside that setting, we evaluate 200 real-image tasks from the Large Vocabulary Instance Segmentation (LVIS) validation split, whose images are from COCO Gupta et al. [2019], Lin et al. [2014]. We evaluate both methods on a valid set of 200 real-image tasks that can be consistently recognized by both detectors.

Table 5 shows that the ordering holds. CITE raises TES from 0.400 to 0.630 and PP from 0.137 to 0.375, and wins 133 of 200 dual-detector comparisons against 31 Baseline wins and

Method	Mask2Former		Mask R-CNN		Average		CITE W/T/L
	TES $\uparrow$	PP $\uparrow$	TES $\uparrow$	PP $\uparrow$	TES $\uparrow$	PP $\uparrow$	
Baseline	0.395	0.150	0.405	0.123	0.400	0.137	133/36/31
CITE	0.640	0.416	0.620	0.333	0.630	0.375	

Table 5: Real-image evaluation on 200 LVIS tasks.

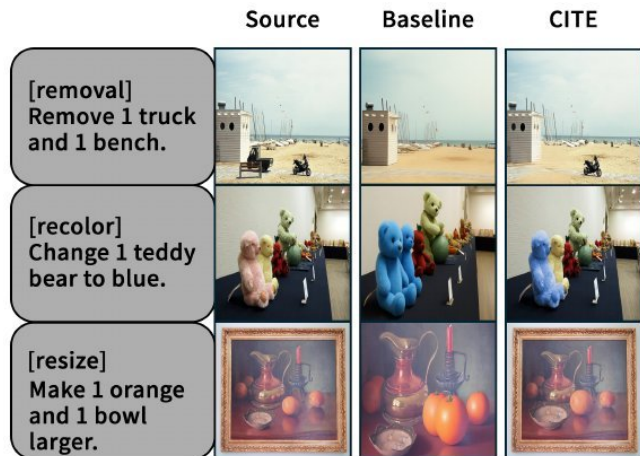


Figure 5: Representative comparison on real LVIS images.

36 ties. Figure 5 shows representative real-image examples, illustrating the same trend qualitatively.

4.8 Human Evaluation

We conduct a blind A/B evaluation on 150 tasks balanced over the three operations and two determinacy states. Each task shows the source image, raw instruction, and randomized Baseline/CITE outputs. Eight raters select the result that edits exactly the requested instances while preserving all other objects. Table 6 shows 788 CITE wins, 95 Baseline wins, and

Split	N	CITE	Tie	Baseline
Removal	400	266	110	24
Recolor	400	259	105	36
Resize	400	263	102	35
Overall	1200	788	317	95

Table 6: Blind human preference by operation. CITE is selected in 89.2% of decisive comparisons.

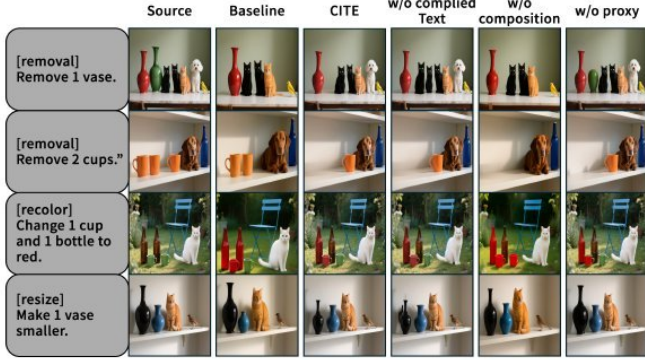


Figure 6: Qualitative component ablation. The proxy condition specifies the edited instances, while latent composition preserves regions outside the edit mask.

317 ties, giving a 78.9% tie-adjusted preference and 89.2% preference among decisive judgments. Inter-rater agreement is moderate for this three-way task (Fleiss’ $\kappa = 0.304$; mean pairwise Cohen’s $\kappa = 0.301$).

4.9 Ablation Study

We ablate the proxy condition, compiled text, and per-step source-latent composition on a 300-task subset. Table 7 and Figure 6 show that the proxy drives target execution: removing it sharply lowers TES. Removing latent composition mainly reduces PP, while removing compiled text has a smaller effect.

4.10 Comparison with Existing Editors

We compare CITE with Blended Latent Diffusion (BLD) Avrahami et al. [2023] and DiffEdit Couairon et al. [2023] on 200 tasks. Table 8 shows higher TES and PP for CITE under both detectors. Figure 7 shows the corresponding instance-level behavior.

5 Conclusion

We introduced CITE, a training-free framework that maps class-level edit requests to target and protect instances. By combining determinacy-aware selection, proxy conditioning, compiled text, and source-latent composition, CITE improves instance-level control and preservation across removal, recoloring, resizing, real images, prompt baselines, and existing editors. CITE remains sensitive to mask quality in crowded

Variant	Determinacy	Mask2Former		Mask R-CNN	
		TES $\uparrow$	PP $\uparrow$	TES $\uparrow$	PP $\uparrow$
Full CITE	unique under	0.964	0.932	0.972	0.940
		0.966	0.950	0.946	0.945
w/o Proxy Condition	unique under	0.594	0.919	0.603	0.921
		0.466	0.939	0.480	0.929
w/o Compiled Text	unique under	0.957	0.931	0.972	0.943
		0.966	0.930	0.966	0.925
w/o Latent Composition	unique under	0.862	0.726	0.901	0.685
		0.466	0.633	0.439	0.610

Table 7: Quantitative component ablation on the same 300-task subset under two independent detector evaluations.

Method	Mask2Former		Mask R-CNN	
	TES $\uparrow$	PP $\uparrow$	TES $\uparrow$	PP $\uparrow$
CITE	0.935	0.940	0.970	0.933
BLD †	0.310	0.018	0.310	0.017
DiffEdit †	0.265	0.030	0.240	0.025

Table 8: Existing-editor comparison on 200 tasks. † Baselines receive CITE’s oracle masks.

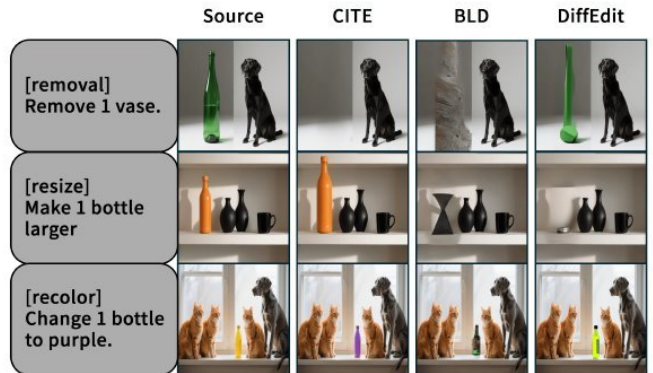


Figure 7: Representative comparison with existing editors.

scenes, thin structures, and shadows; future work will improve support estimation and extend operations.

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